

Wavefronts for degenerate diffusion-convection reaction equations with sign-changing diffusivity

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Diffusion-convection reaction equations

We deal with the following parabolic equation in one spatial dimension:

$$\rho_t + f(\rho)_x = \{D(\rho)\rho_x\}_x + g(\rho) \quad \text{for } x \in \mathbb{R}, \quad t \geq 0. \quad (1)$$

- The unknown is $\rho = \rho(x, t) \in [0, 1]$. (In view of applications ρ is to be understood as a (normalized) **density**.)
- f , D and g are **density-dependent** and such that:
 - (f) $f \in C^1[0, 1]$ with $f(0) = 0$; (**convection**)
 - $D \in C^1[0, 1]$ that **changes sign** in $(0, 1)$; (**diffusion**)
 - (g) $g \in C^0[0, 1]$ with $g > 0$ in $(0, 1)$ and $g(0) = g(1) = 0$. (**reaction of monostable-type**)

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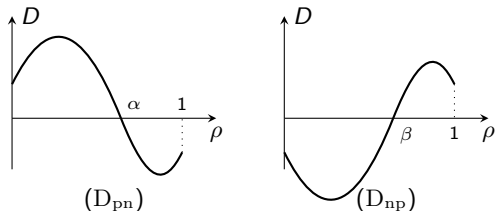
The zeros of g are the equilibria of (1): Setting $\rho(x, t) = \ell$, where $\ell = 0, 1$, gives a constant solution of (1).

Solutions of (1) are determined up to spatial shifts: If ρ is a solution then $(t, x) \mapsto \rho(t, x - x_0)$ is still a solution.

The main assumption is on the diffusivity D which **changes sign** once in $(0, 1)$, namely for some $\alpha, \beta \in (0, 1)$:

(D_{pn}) $D \in C^1[0, 1]$, $D > 0$ in $(0, \alpha)$ and $D < 0$ in $(\alpha, 1)$;

(D_{np}) $D \in C^1[0, 1]$, $D < 0$ in $(0, \beta)$ and $D > 0$ in $(\beta, 1)$.



Equation (1) is a **forward** parabolic equation where $D > 0$ and a **backward** parabolic equation where $D < 0$.

D can **vanish** at one (or both) extremum of $[0, 1]$.

Wavefronts

Traveling-wave solutions (TWs) are solutions such that

$$\rho(x, t) = \varphi(x - ct),$$

for some wave profile $\varphi = \varphi(\xi)$ and a wave speed $c \in \mathbb{R}$. **Wavefronts** are TWs whose profile is *globally-defined*, *non-constant* and *monotone (non-increasing)*.

In looking for wavefronts of (1), we are then interested in profiles φ satisfying, for some $c \in \mathbb{R}$,

$$\left(D(\varphi) \varphi' \right)' + (c - h(\varphi)) \varphi' + g(\varphi) = 0 \quad \text{in } \mathbb{R}, \quad (2)$$

where $h = \dot{f}$.

(Note: we look for φ continuous, a.e. differentiable such that $D(\varphi)\varphi' \in L^1_{\text{loc}}$ satisfying (2) in **distributional sense**.)

Because of (g) , we couple (2) with the (infinite) boundary condition

$$\varphi(-\infty) = 1 \quad \text{and} \quad \varphi(+\infty) = 0. \quad (3)$$

About TWs for (1)

Regarding the case $D \geq 0$, the seminal work of [Aronson-Weinberger \(1978\)](#) studied the case $D = 1$ and $f = 0$. A more detailed study is due to [Marcelli-Malaguti](#) (with $f \neq 0$) and [Gilding-Kersner](#) (mainly when either $f = 0$ or $g = 0$) in the [early 2000's](#).

The case of $\text{sign-changing } D$ has been studied recently ([Maini-Malaguti-Marcelli-Matucci \(2006\)](#), [Bao-Zhao \(2014\)](#)) but only in the case $f = 0$.

Typical existence result

Profiles exist if and only if $c \geq c^*$, for a real threshold $c^* \in \mathbb{R}$.

Major issues are also:

- The occurrence of profiles that reach an equilibrium for a **finite value** ξ^* .
- The regularity of φ where D degenerates: φ can be **classical** or **sharp**.
(φ is sharp at ℓ if there exists $\xi_\ell \in I$ such that $\varphi(\xi_\ell) = \ell$, with φ classical in $I \setminus \{\xi_\ell\}$ and not differentiable at ξ_ℓ .)

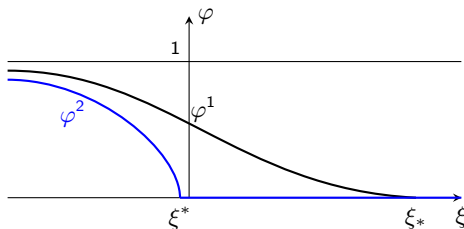


Figure: $D(0) = 0$: φ^1 reaches 0 at some finite value ξ_* in a differentiable way while φ^2 is sharp at 0.

Remark. φ sharp at 0 (or 1) implies that φ reaches 0 (or 1) at a finite value $\xi^* \in \mathbb{R}$.

Our contributions

In collaboration with Corli (Ferrara) and Malaguti (Modena and Reggio Emilia), we have provided the existence (uniqueness), regularity and (strictly) monotonicity of wavefronts for Equation (1) under (f) , $(D_{pn}) - (D_{np})$ and (g) , highlighting the convective effect.

The results are contained in

- * [B.-Corli-Malaguti, (submitted to) Discrete Contin. Dyn. Syst. 2021]

and are based also on properties of suitable semi-wavefronts obtained in

- * [B.-Corli-Malaguti, Electron. J. Qual. Theory Differ. Equ. 2020].

A study on TWs for (1) with *sign-changing diffusivity* and *bistable (or Allen-Cahn) reaction* is contained in:

- * [B.-Corli-Malaguti, in preparation].

Case (D_{pn}): positive-to-negative diffusivity

We need to assume

$$\limsup_{\varphi \rightarrow 0^+} \frac{D(\varphi)g(\varphi)}{\varphi} < +\infty \quad \text{and} \quad \limsup_{\varphi \rightarrow 1^-} \frac{D(\varphi)g(\varphi)}{\varphi - 1} < +\infty. \quad (4)$$

Theorem 1 ([Theorem 2.1, B.-Corli-Malaguti (2021)])

Assume (f), (D_{pn}), (g) and (4). Equation (1) admits a (unique up to space shifts) wavefront, with speed c and profile φ satisfying (3), if and only if $c \geq c_{pn}^*$, for some $c_{pn}^* \in \mathbb{R}$.

For such c , we have $\varphi'(\xi) < 0$ when $\varphi(\xi) \in (0, 1) \setminus \{\alpha\}$; there exists a unique $\xi_\alpha \in \mathbb{R}$ such that $\varphi(\xi_\alpha) = \alpha$ and

$$\varphi'(\xi_\alpha^\pm) = \begin{cases} < 0 & \text{if } \dot{D}(\alpha) < 0 \text{ or } c > h(\alpha), \\ -\infty & \text{if } \dot{D}(\alpha) = 0 \text{ and } c \leq h(\alpha). \end{cases}$$

Theorem 2 ([Theorem 2.1, B.-Corli-Malaguti (2021)])

Under the assumptions of Theorem 1, let φ be a profile of a wavefront given by Theorem 1. It holds that:

- if $D(0)D(1) < 0$, then φ is strictly decreasing and hence classical in $\mathbb{R} \setminus \{\xi_\alpha\}$;
- if $D(0)D(1) = 0$ and $c > c_{pn}^*$, then φ is classical in $\mathbb{R} \setminus \{\xi_\alpha\}$;
- if $D(0) = 0$ and $c = c_{pn}^* = c_{p,r}^* > h(0)$, then φ is sharp at 0 (reached at some $\xi_0 > \xi_\alpha$);
- if $D(1) = 0$ and $c = c_{pn}^* = c_{n,l}^* > h(1)$, then φ is sharp at 1 (reached at some $\xi_1 < \xi_\alpha$).

– From [BCM (2020)], φ is strictly decreasing if

$$c > \max \left\{ h(0) + \limsup_{\varphi \rightarrow 0^+} \frac{g(\varphi)}{\varphi}, h(1) + \limsup_{\varphi \rightarrow 1^-} \frac{g(\varphi)}{1 - \varphi} \right\};$$

– Explicit expression for φ' at ξ_0 and ξ_1 are given.

Comments on Theorems 1-2

We have:

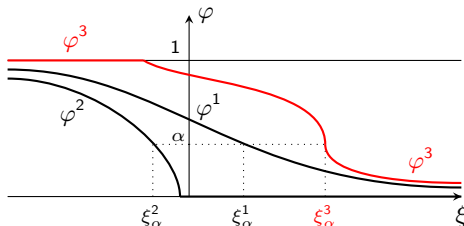
- Theorems 1 and 2 have the merit to extend previous results known in the case $f = 0$ ([Maini-Malaguti-Marcelli-Matucci (2006)]).
- The presence of f implies that, for every $c \geq c_{pn}^*$, we can have

$$\varphi'(\xi_\alpha) = -\infty,$$

while if $f = 0$ profiles are of class C^1 .

- Based on [BCM (2020)], Theorem 1 inherits precise estimates on c_{pn}^* . Essentially, the addition of f shifts the bounds for c_{pn}^* .

Some possible wavefronts joining 1 with 0 in case (D_{pn}) .



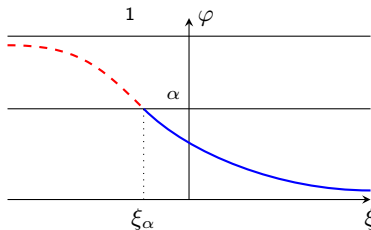
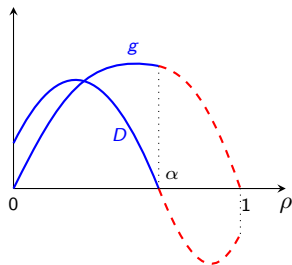
- a classical wavefront $\varphi^1 : c > c_{pn}^*$ and either $\dot{D}(\alpha) < 0$ or $c > h(\alpha)$;
- $\varphi^2 : c = c_{p,r}^* > h(0)$, $D(0) = 0 = \dot{D}(0)$;
- φ^3 with $(\varphi^3)'(\xi_\alpha^3) = -\infty : \dot{D}(\alpha) = 0$ and $c \leq h(\alpha)$.

Proof of Theorem 1 (an outline)

To build up wavefronts of (1), our approach relies on **properly joining two semi-wavefronts**, each of them with intrinsic thresholds.

A **semi-wavefronts to 0**, connecting α to 0 is a TW whose profile is defined in $(\xi_\alpha, +\infty)$, non-increasing, non-constant, such that $\varphi(+\infty) = 0$ and such that $\varphi(\xi_\alpha^+) = \alpha$. A semi-wavefront to 0 is said **strict** if $\xi_\alpha \in \mathbb{R}$.

Analogously, we define a **strict semi-wavefront from 1**, connecting 1 to α .



Existence of semi-wavefronts

Consider the singular first-order problem

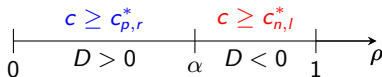
$$\begin{cases} \dot{z}(\varphi) = h(\varphi) - c - \frac{D(\varphi)g(\varphi)}{z(\varphi)} & \text{for } \varphi \in (0, \alpha), \\ z(\varphi) < 0 & \text{for } \varphi \in (0, \alpha), \\ z(0^+) = 0, \end{cases} \quad (5)$$

deduced by the the singular change of variable

$$z(\varphi) := D(\varphi)\varphi', \quad \text{for } 0 < \varphi < \alpha. \quad (6)$$

In [Theorem 2.2, BCM (2020)], we proved that (1) admits **strict semi-wavefronts to 0, connecting α to 0**, with speed c if and only if (5) is solvable, that is if and only if $c \geq c_{p,r}^*$, for a certain $c_{p,r}^* \in \mathbb{R}$.

From [Proposition 3.2, BCM (2021)], (1) admits **strict semi-wavefronts from 1, connecting 1 to α** , with speed c if and only if $c \geq c_{n,l}^*$, for a $c_{n,l}^* \in \mathbb{R}$.



Define

$$c_{pn}^* := \max \{ c_{p,r}^*, c_{n,l}^* \}.$$

On the uniqueness of the semi-wavefronts pasting

- In [B-Corli-Malaguti (2020)] it was shown a **one-parameter family** of semi-wavefronts to 0, connecting α to 0, parameterized by the value

$$\lim_{\xi \rightarrow \xi_\alpha^+} (D(\varphi)\varphi')(\xi) = z(\alpha^-).$$

- Nonetheless, to construct a wavefront as in Theorem 1, **we can benefit only from one profile** from that family. More precisely, from the profile such that the associated function z satisfies

$$z(\alpha^-) = 0.$$

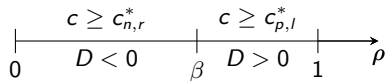
- In all the other cases, the pasting of profiles does not provide a solution (according to the distributional sense) in a neighborhood of the matching point.

Case (D_{np}): negative-to-positive diffusivity

Let c_{np}^* be defined by

$$c_{np}^* := \max \{ c_{n,r}^*, c_{p,l}^* \},$$

where $c_{n,r}^*$ and $c_{p,l}^*$ are the thresholds for which semi-wavefronts connecting (resp.) β to 0 and 1 to β exist.



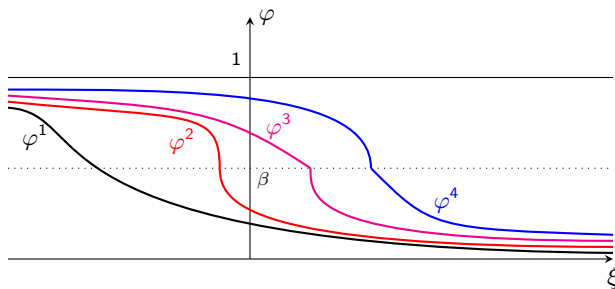
Theorem 3 ([Theorem 2.2, B.-Corli-Malaguti (2021)])

Assume (f), (D_{np}) and (g). Then, Equation (1) admits a (unique up to space shifts) wavefront, with speed c and profile φ satisfying (3), if and only if $c \geq c_{np}^*$.

For such c , we have $\varphi' < 0$ if $\varphi \in (0, 1) \setminus \{\beta\}$. There exists a unique $\xi_\beta \in \mathbb{R}$ such that $\varphi(\xi_\beta) = \beta$ and

- ❶ $\varphi'(\xi_\beta) \in \mathbb{R}$ if $c > c_{np}^*$;
- ❷ $\varphi(\xi_\beta^+) = \varphi'(\xi_\beta^-)$ (possibly not finite) if $c = c_{np}^* = c_{n,r}^* = c_{p,l}^*$;
- ❸ $\varphi'(\xi_\beta^-) > \varphi'(\xi_\beta^+)$ if $c = c_{np}^* = c_{n,r}^* > c_{p,l}^*$;
- ❹ $\varphi'(\xi_\beta^-) < \varphi'(\xi_\beta^+)$ if $c = c_{np}^* = c_{p,l}^* > c_{n,r}^*$.

Comments on Theorem 3



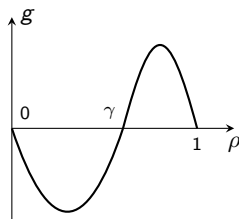
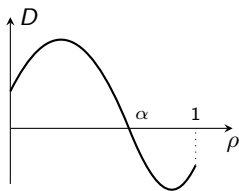
Mainly:

- Theorem 3 extends results already known in the case $f = 0$ ([Bao-Zhou (2014)]).
- The presence of f reveals profiles only continuous at ξ_β , **in contrast with the case $f = 0$.**
- Due to the presence of f , if $D(0)D(1) = 0$, we prove the existence of possible sharp profiles at either 0 and 1. (**Not possible when $f = 0$.**)
- Fine estimates on c_{np}^* .

Forward-backward equations with Allen-Cahn reaction term

We deal now with (1) with, for some $\alpha, \gamma \in (0, 1)$,

- $D \in C^1[0, 1]$, $D > 0$ in $(0, \alpha)$ and $D < 0$ in $(\alpha, 1)$;
- $g \in C^0[0, 1]$, $g < 0$ in $(0, \gamma)$, $g > 0$ in $(\gamma, 1)$ and $g(0) = g(1) = 0$.



If $f = 0$ then wavefronts exist at most for a unique speed $c > 0$ and only if $\alpha > \gamma$.

In [BCM (in preparation)], we prove that the presence of the convective term f produces **completely new phenomena**: solutions exist regardless the relative order between α and γ and if $\alpha \leq \gamma$ a whole interval of speeds can be admitted and uniqueness (up to shifts) is lost.

Thanks for your attention!

A formal definition

We define now traveling waves and the related profiles. We denote with $I \subseteq \mathbb{R}$ an open interval. We refer to [Gilding-Kersner \(2004\)](#).

Definition of TWs

Let c be a real constant and φ be a function such that

- $\varphi \in C(I)$ with values in $[0, 1]$, for some $I \subset \mathbb{R}$;
- φ is differentiable a.e.;
- $D(\varphi)\varphi' \in L^1_{\text{loc}}(I)$.

The function $\rho(x, t) := \varphi(x - ct)$, for (x, t) with $x - ct \in I$, is a **traveling-wave** solution of equation (1) with wave speed c and wave profile φ if, for every $\psi \in C_0^\infty(I)$,

$$\int_I \left(D(\varphi(\xi)) \varphi'(\xi) - f(\varphi(\xi)) + c\varphi(\xi) \right) \psi'(\xi) - g(\varphi(\xi)) \psi(\xi) d\xi = 0.$$

A TW solution is

global if $I = \mathbb{R}$ and **strict** if $I \neq \mathbb{R}$ and φ is not extendible to $\mathbb{R}$;

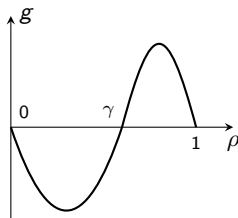
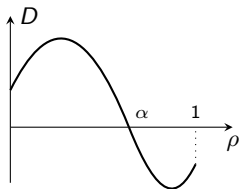
classical if φ is differentiable, $D(\varphi)\varphi'$ is absolutely continuous and (2) holds a.e.;

sharp at ℓ if there exists $\xi_\ell \in I$ such that $\varphi(\xi_\ell) = \ell$, with φ classical in $I \setminus \{\xi_\ell\}$ and not differentiable at ξ_ℓ ;

a **wavefront** if it is global, with a monotonic, non-constant profile φ satisfying either (3) or the converse condition;

a **semi-wavefront to 1** (or *to 0*) if $I = (a, \infty)$ for $a \in \mathbb{R}$, the profile φ is monotonic, non-constant and $\varphi(\xi) \rightarrow 1$ (respectively, $\varphi(\xi) \rightarrow 0$) as $\xi \rightarrow \infty$; a **semi-wavefront from 1** (or *from 0*) if $I = (-\infty, b)$ for $b \in \mathbb{R}$, the profile φ is monotonic, non-constant and $\varphi(\xi) \rightarrow 1$ (respectively, $\varphi(\xi) \rightarrow 0$) as $\xi \rightarrow -\infty$.

Forward-backward equations with Allen-Cahn reaction term



In [Maini-Malaguti-Marcelli-Matucci (2007)] was considered the case $f = 0$. The authors prove that

$$\alpha > \gamma, \quad c > 0 \quad \text{and} \quad \int_0^\alpha D(\rho)g(\rho) d\rho > 0$$

are needed in order to have solutions.

Moreover, under further assumptions they prove that there exists a **unique admissible** $c^* > 0$ for which (1) admits a (unique up to shifts) wavefront.

In [BCM3], we prove that the presence of the convective term f produces **completely new phenomena**.

Not only either the appearance of less-regular solutions and shifts in the thresholds are admitted but, more intimately, a large variety of new solutions and admissible speeds are also proved to hold.

$f \neq 0 \implies$ (possible) solutions even if $\alpha = \gamma$ or $\alpha < \gamma$.

if $\alpha \leq \gamma \implies$ a whole interval of admissible speeds.

if $\alpha = \gamma \implies$ (possible) plateau profiles.

if $\alpha < \gamma \implies$ (possible) plateau profiles; for every c , a family of profiles.

On semi-wavefronts to 0

The main result about semi-wavefronts is the following.

Theorem 4 ([Theorem 2.2, B.-Corli-Malaguti (2020)])

Assume (f), (D1), (g0) and (4)₁. Then, there exists $c^ \in \mathbb{R}$, which satisfies*

$$\max \left\{ \sup_{(0,1]} \frac{f(\varphi)}{\varphi}, h(0) + 2\sqrt{\liminf_{\varphi \rightarrow 0^+} \frac{D(\varphi)g(\varphi)}{\varphi}} \right\} \leq c^* \leq 2\sqrt{\sup_{(0,1]} \frac{D(\varphi)g(\varphi)}{\varphi}} + \sup_{(0,1]} \frac{f(\varphi)}{\varphi},$$

such that (1) has strict semi-wavefronts to 0, connecting 1 to 0, if and only if $c \geq c^$. Moreover, if φ is the profile of one of such semi-wavefronts, then it holds that*

$$\varphi'(\xi) < 0 \quad \text{for any } 0 < \varphi(\xi) < 1.$$

Key tool to deal with semi-wavefronts (to 0)

The main issue in solving (here for $\varphi \in (0, \alpha)$)

$$\begin{cases} (D(\varphi) \varphi')' + (c - h(\varphi)) \varphi' + g(\varphi) = 0, \\ \varphi(+\infty) = 0, \end{cases} \quad (7)$$

by a classical dynamical system analysis is the lack of regularity of the coefficients (D , g and h). Hence we need to investigate existence and regularity of fronts with a different method.

Our main tool is the **reduction** of (7) to the singular first-order problem

$$\begin{cases} \dot{z}(\varphi) = h(\varphi) - c - \frac{D(\varphi)g(\varphi)}{z(\varphi)} & \varphi \in (0, \alpha), \\ z(\varphi) < 0 & \varphi \in (0, \alpha), \\ z(0) = 0. \end{cases} \quad (8)$$

The above problem is deduced by the singular change of variable

$$z(\varphi) := D(\varphi) \varphi',$$

where the right-hand side is understood to be computed at $\varphi^{-1}(\varphi)$.